RAW SOURCE	PREPARATION	DERIVATIVE TABLE	REFERENCE SET
GSHP (global) Gupta et al. 2022 15,259 layers params + raw points	`prepare_gshp.py` keep “good quality estimate”, valid class, fractions 95–105 %; Australian rows out α m^{-1} → kPa^{-1} · K_s cm/d → cm/h	`gshp_reference` **9,900** 6,552 with K_s	`merged` (default) **21,720** layers 11,246 carry a measured saturated conductivity 7,862 European layers (900 + 483 + 6,479) 12 USDA classes; sand is 21 % of rows, silt under 0.4 % Without the restricted EU-HYDI and Boorowa tables, merged: the 14,866 (--reference public)
NCSS / KSSL (USA) USDA–NRCS, 5.4 GB ~420,000 layers gravimetric %, no K_s	`prepare_kssl.py` need ≥5 tensions + particle-size analysis w% × bulk density → volumetric θ add saturation anchor θ(0) = 0.95 × porosity	`kssl_reference` **2,530** 0 with K_s	
Hohenbrink (Germany) et al. 2023, ESSD 572 German cores HYPROP + WP4 + KSAT	`prepare_hohenbrink.py` pF → kPa · k cm/d → cm/h bin in pF, one median point per bin (else 199 wet points outvote 3 dry ones)	`hohenbrink_reference` **560** 402 with K_s	
EU-HYDI (Europe) Weynants et al. 2013 18,682 samples, 18 countries consortium-restricted	`prepare_euhydi.py` need ≥5 reliable points + USDA fractions cm → kPa · K at h = 0, saturated methods only bin in pF; sum raw size classes where empty	`euhydi_reference` **6,797** 2,704 with K_s · not distributed	
Zanjanrood (Iran) Babaeian et al. 2015 173 samples, 0–30 cm cores + sieved dry end	`prepare_babaeian_zanjanrood.py` refit vG to measured points, cm → kPa need a point wetter than 330 cm K_s on repacked samples, used as intact	`babaeian_zanjanrood_reference` **169** 169 with K_s	
Laikipia (Kenya) Willard et al. 2026, HydroShare 70 cores, 35 fields the project’s own data	`prepare_willard.py` published curves (HydroShare) density follows shrinkage, cap 1.6 16 dispersed samples left out	`willard_reference` **70** 26 with K_s (Saturo)	
Arizona (USA) E. Babaeian, shared data 21 samples, sand to clay no coordinates	`prepare_babaeian_az.py` refit vG to points up to 1500 kPa interpolate a low 500 cm reading (3 samples) the file's own vG fits not used	`babaeian_az_reference` **21** 21 with K_s	
Canary Islands (Spain) Armas Espinel 2013 66 cores, andic soils thesis, Tenerife	`prepare_armas.py` points read from the thesis's vector figures checked against its tables and vG fits resin-dispersed fractions kept alongside	`armas_reference` **66** 62 with K_s	
Yellow River (China) Tong et al. 2024, PANGAEA 1,035 curves, 475 sites centrifuge; laser texture	`prepare_tong.py` only fits published: rebuild 1–1000 kPa add measured θs at h = 0, refit α m^{-1} → kPa^{-1} · K_s cm/min → cm/h	`tong_reference` **1,030** 1,015 with K_s	
Andalusia (Spain) K. Vanderlinden, IFAPA 122 curves, four sites distributed	`prepare_andalusia.py` gravimetric × ring density (Setenil) Tomejil, Lebrija: density follows shrinkage points to 1500 kPa; Doñana sands	`setenil+tomejil+lebrija+donana` **122** 44 with K_s	
CSIRO (Australia) NatSoil API + Boorowa Farm 224 + 57 cores CC BY 4.0 / BY-NC-SA	`prepare_natsoil.py` texture to USDA 50 μm limit (Sedigraph, sieve, or interpolated); coordinates fixed replaces GSHP’s Australian rows	`natsoil+boorowa_reference` **281** 105 with K_s · Boorowa local	
Japan, SE Asia Fujii et al. 2023, Dryad 59 cores, 23 forest/farm sites CC0 public domain	`prepare_fujii.py` points to pF 4.2; pF 5.5 and 7 dropped sheets joined on site and depth class re-derived from the fractions	`fujii_reference` **59** 51 with K_s	
Ksat-SSIR-DB (Brazil) Ottoni et al. 2025, SGB 2,579 rows; 215 with a curve public database	`prepare_ksatssir.py` all five points: 0, 6, 10, 33, 1500 kPa 100 rows at HYBRAS v1 sites left out lab K_s on rings only, mm/h → cm/h	`ksatssir_reference` **115** 95 with K_s	
UNSODA 2.0 (worldwide) 790 soils	`prepare_unsoda.py` fit vG to measured points, cm → kPa	`unsoda_reference` **581**	`all` **22,466** adds UNSODA and sDB; both largely duplicated inside GSHP already, so not in the default
sDB (Belgium) PANGAEA, CC-BY-3.0	`prepare_sdb.py` fit vG to measured points	`sdb_reference` **165**	

`free_m.py` — shared by every preparation script

Also writes an unconstrained companion fit (θr, θs, α, n, m) alongside the Mualem one, refitting measured points where a source ships them.